Timeline Check-in

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Set up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)
library(ggridges)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")
```

```r
# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning and Wrangling

Cleaning

`sites` object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

`taxon_list` object

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake")) |>
  select(verbatim_name, scientificName)
```

`water_quality` object

```r
water_quality_clean <- water_quality |>

  # cleaning column names
  clean_names() |>

   # filtering join: only include sites in ncos_sites
```

```
  # keep all the rows on the left that match up with rows on the right
  semi_join(ncos_sites, by = "site") |>

  # combine date and site together and remove those indiv. columns
  unite("date_site", date, site, remove = TRUE) |>

  # combine date and site together and remove those indiv. columns
  group_by(date_site) |>

  # create new columns that show median of each variable at each date_site
  # because there are multiple samples at the same site at different depths
  summarize(med_ph    = median(p_h, na.rm = TRUE),
            med_sal   = median(salinity_ppt, na.rm = TRUE),
            med_do    = median(dissolved_oxygen_mg_l, na.rm = TRUE),
            med_temp  = median(temperature_c, na.rm = TRUE)) |>

  # ungroup
  ungroup() |>

  # take out outlier
  filter(date_site != "2022-08-12_NVBR")
```

`aq_ins` object

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         cladocera) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:cladocera,
```

```r
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
 ↪  liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~  "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
 ↪  = TRUE),
       site_full = fct_rev(site_full)) |>

# log+1 transform of average abundance per liter
mutate(log_ave_lit = log(ave_lit + 1)) |>                        # <-- Fig
 ↪  1, 2, 3, 4

# season derived from date
mutate(season = case_when(
  month(date) %in% c(3, 4, 5)  ~ "Spring",
  month(date) %in% c(6, 7, 8)  ~ "Summer",
  month(date) %in% c(9, 10, 11) ~ "Fall",
  month(date) %in% c(12, 1, 2) ~ "Winter"
),
season = factor(season, levels = c("Spring", "Summer", "Fall", "Winter")))
 ↪  |># <-- Fig 4
```

```
  mutate(taxon = factor(taxon, levels = c("copepod", "ostracod",
   ↪  "cladocera")))

# capitalizing invertebrate names
aq_ins_clean$taxon <- str_to_sentence(aq_ins_clean$taxon)
```

Setting plot aesthetics

```
# colors for taxa
taxon_cols <- c(
  "Ostracod" = "orange2",
  "Copepod" = "tomato3",
  "Cladocera" = "turquoise4")

sites_cols <- c(
  "NDC" = "hotpink3",
  "NEC" = "cyan3",
  "NMC" = "lightpink2",
  "NPB" = "slateblue3",
  "NPB1" = "steelblue3",
  "NPB2" = "green3",
  "NVBR" = "maroon3"

)

# setting common theme elements
theme_set(
  theme_classic() +
    theme(text = element_text(size = 12),
          plot.title.position = "plot",
          legend.position = "none"))
```

Visualizations directly relevant to answering questions

Figure 1

```
# base layer: create new object to store figure 1
figure1 <- ggplot(data = aq_ins_clean,
```

```
                  mapping = aes(x = med_temp,
                                y = log_ave_lit,
                                color = taxon)) +
  # first layer: points
  geom_point() +

  # facet by site
  facet_wrap(~site_full) +

  # labels
  labs(title = "Figure 1:

Dominant Three Taxa Abundance and Average Water Temperature by Site",
       x = "Average Temperature (ºC)",
       y = "log + 1 Transformed Average Abundance/L",
       color = "Taxon") +

  # manual colors
  scale_color_manual(values = taxon_cols) +

  # adding legend
  theme(legend.position = "right")



figure1
```

Figure 1:

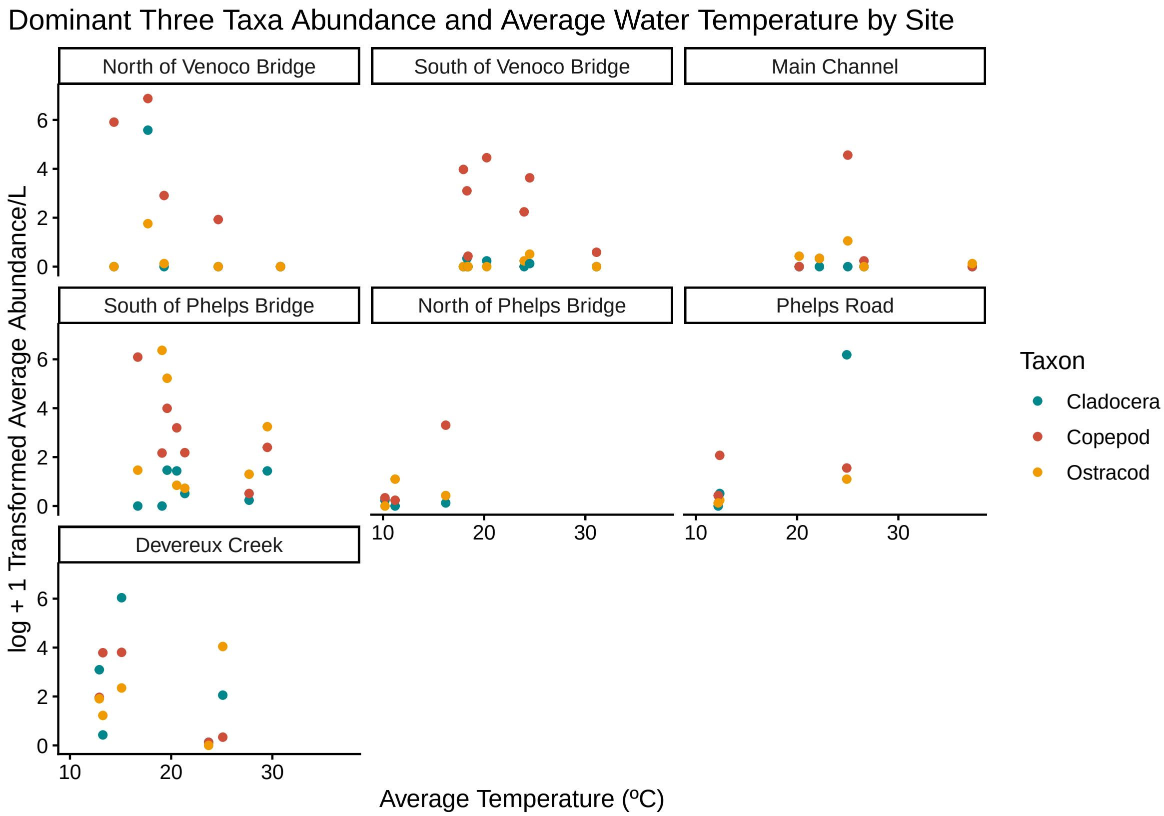


Figure 2

```
# base layer: create new object to store figure 2
figure2 <- ggplot(data = aq_ins_clean,
                  aes(x = taxon,
                      y = log_ave_lit,
                      color = taxon)) +

  # first layer: jitter plot
  geom_jitter(height = 0,
              alpha = 1,
              width = 0.2,
              shape = 21) +

  # second layer: points to represent medians
```

```
  stat_summary(geom = "point",
               fun = "mean",
               size = 4,
               color = "black",
               shape = 18) + # 18 = filled diamond

   stat_summary(geom = "errorbar",
               fun.data = mean_se,
               width = 0.1,
               color = "black") +

  # colors
  scale_color_manual(values = taxon_cols) +

  # titles
  labs(x = "Taxon",
       y = "log + 1 transformed average abundance/L",
       title = "Figure 2:

Mean and Distribution of Average Abundance for Three Dominant Taxa")

figure2
```

Figure 2:

Mean and Distribution of Average Abundance for Three Dominant Taxa

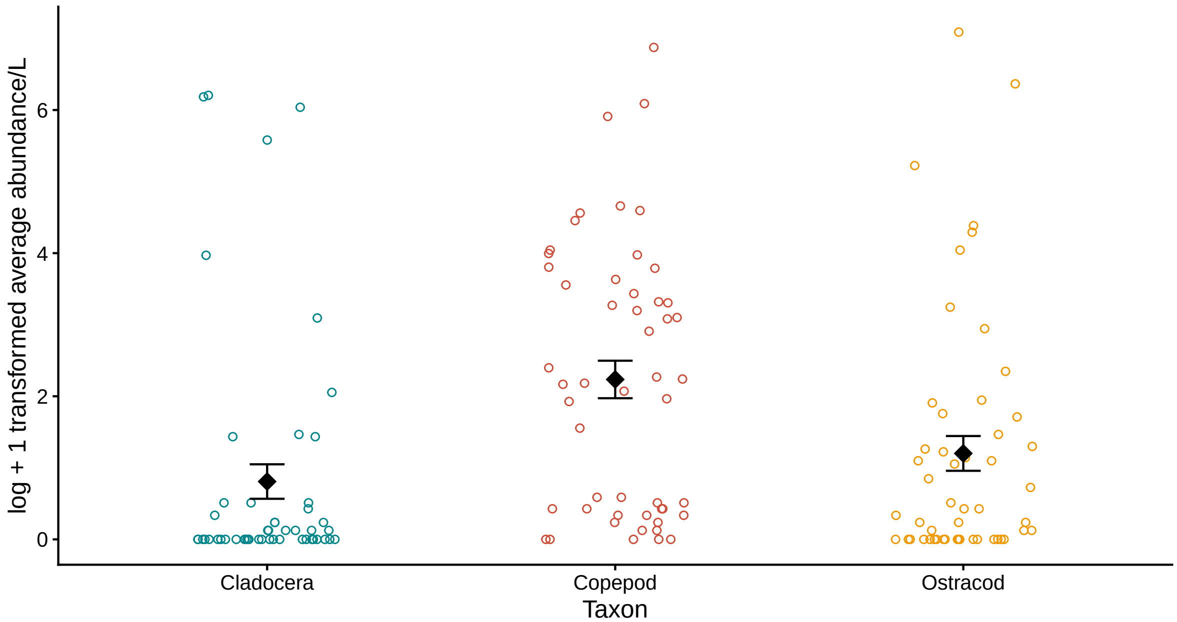
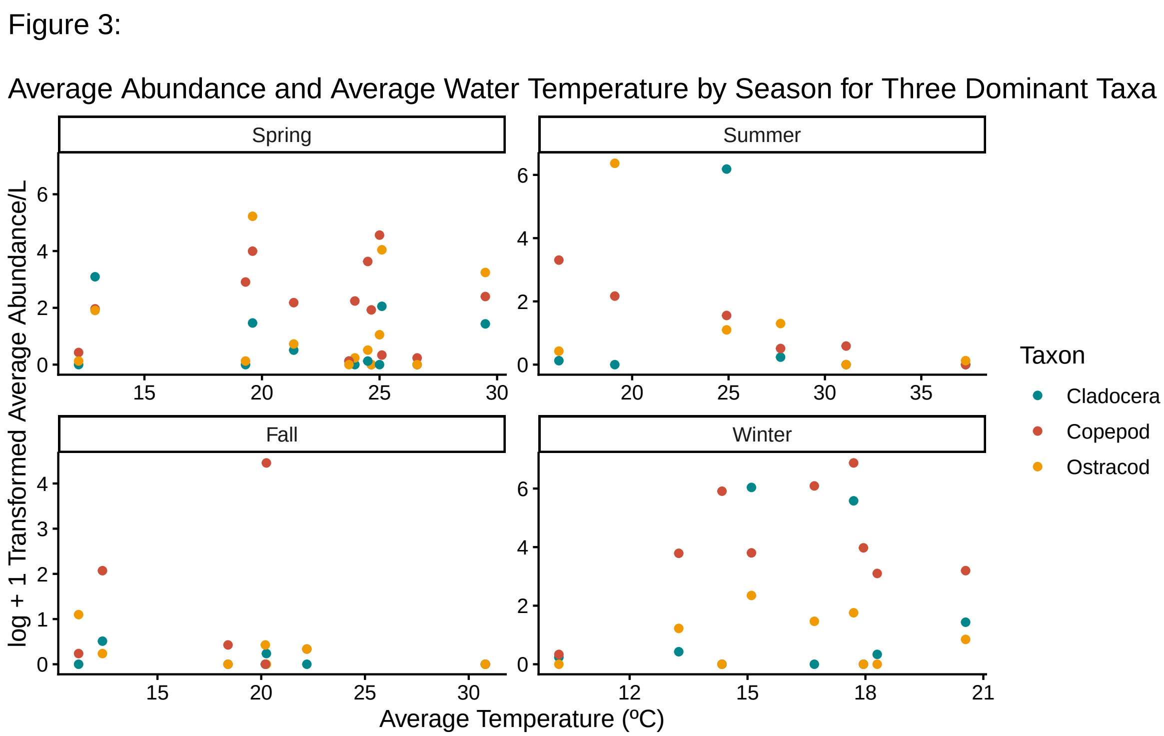


Figure 3 - What is the strength of the relationship between water temperature and invertebrate abundance, and does this differ by season?

```
# base layer: create new object to store figure 4
figure3 <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = med_temp,
                                y = log_ave_lit,
                                color = taxon)) +
  # first layer: points
  geom_point() +

  # facet by season
  facet_wrap(~season,
             scales = "free") +

  # titles and labels
  labs(title = "Figure 3:

Average Abundance and Average Water Temperature by Season for Three Dominant
 ↪  Taxa",
```

```
      x = "Average Temperature (ºC)",
      y = "log + 1 Transformed Average Abundance/L",
      color = "Taxon") +

  # manual colors
  scale_color_manual(values = taxon_cols) +

  # legend
  theme(legend.position = "right")


figure3
```



Other exploratory visualizations

Figure 4 - How does abundance vary across temperature ranges for each taxon?

```
aq_ins_clean <- aq_ins_clean |>
  mutate(temp_range = cut(med_temp,
                          breaks = c(10.0, 15.6, 21.2, 26.8, 32.4, 38.0),
                          labels = c("10.0-15.6C", "15.6-21.2C",
                           ↪  "21.2-26.8C",
                                     "26.8-32.4C", "32.4-38.0C"),
                          include.lowest = TRUE),
         point_color = if_else(log_ave_lit == 0, "Zero Abundance",
          ↪  as.character(temp_range)),
         point_color = factor(point_color, levels = c("10.0-15.6C",
          ↪  "15.6-21.2C",
                                                       "21.2-26.8C",
                                                        ↪  "26.8-32.4C",
                                                       "32.4-38.0C", "Zero
                                                        ↪  Abundance")))



# base layer: create new object to store figure 3a
figure4 <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = med_temp,
                                y = log_ave_lit,
                                color = temp_range)) +
  # first layer: points
  geom_point() +

  # facet by taxon
  facet_wrap(~taxon,
             scales = "free") +

  # add legend
  theme(legend.position = "right")+

  # labels
  labs(title = "Figure 4:

Average Abundance and Average Water Temperature for Three Dominant Taxa",
       x = "Average Temperature (ºC)",
       y = "log + 1 Transformed Average Abundance/L",
color = "Temperature Range") +

  scale_x_continuous(
```

```
    breaks = c(10.0, 15.6, 21.2, 26.8, 32.4, 38.0),
    limits = c(10.0, 38.0)) +

  scale_color_manual(values = c(
    "10.0-15.6C" = "steelblue",
    "15.6-21.2C" = "skyblue",
    "21.2-26.8C" = "gold",
    "26.8-32.4C" = "orange",
    "32.4-38.0C" = "tomato3",
    "Zero Abundance" = "black"
  ), na.value = "grey50")


figure4
```

Figure 4:

Average Abundance and Average Water Temperature for Three Dominant Taxa

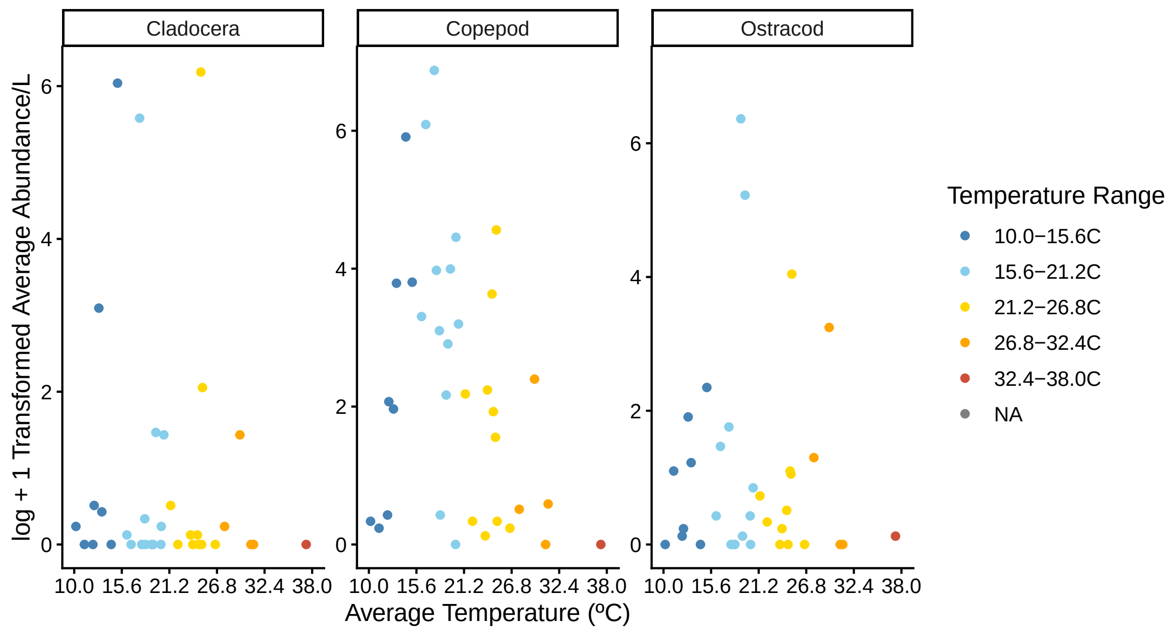


Figure 5 - How does abundance vary across sites for each taxon?

```
# base layer: create new object to store figure 3b
```

```
figure5 <- ggplot(data = aq_ins_clean,
                   mapping = aes(x = site,
                                 y = log_ave_lit,
                                 color = site)) +

  # first layer: points
  geom_point() +

  # facet by taxon
  facet_wrap(~taxon) +

  # add legend
  theme(legend.position = "right") +

  # manual colors
  scale_color_manual(values = sites_cols) +

  # labels and title
  labs(title = "Figure 5:

Average Abundance by Site for Three Dominant Taxa",
       x = "Site",
       y = "log + 1 Transformed Average Abundance/L",
       color = "Site")


figure5
```

Figure 5:

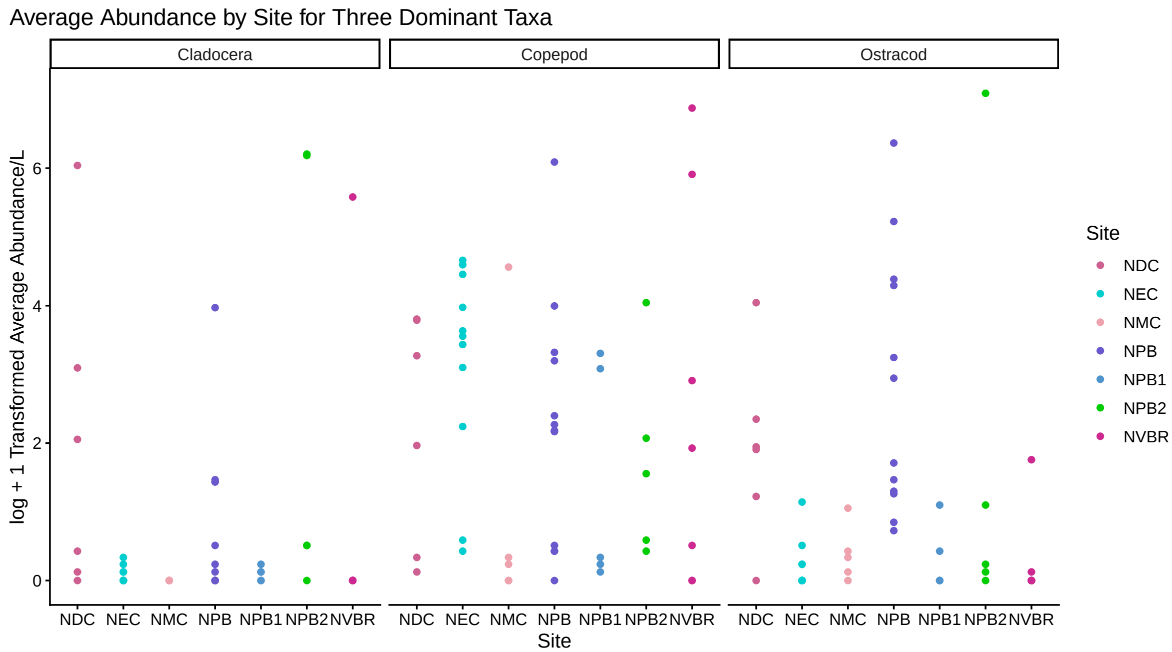
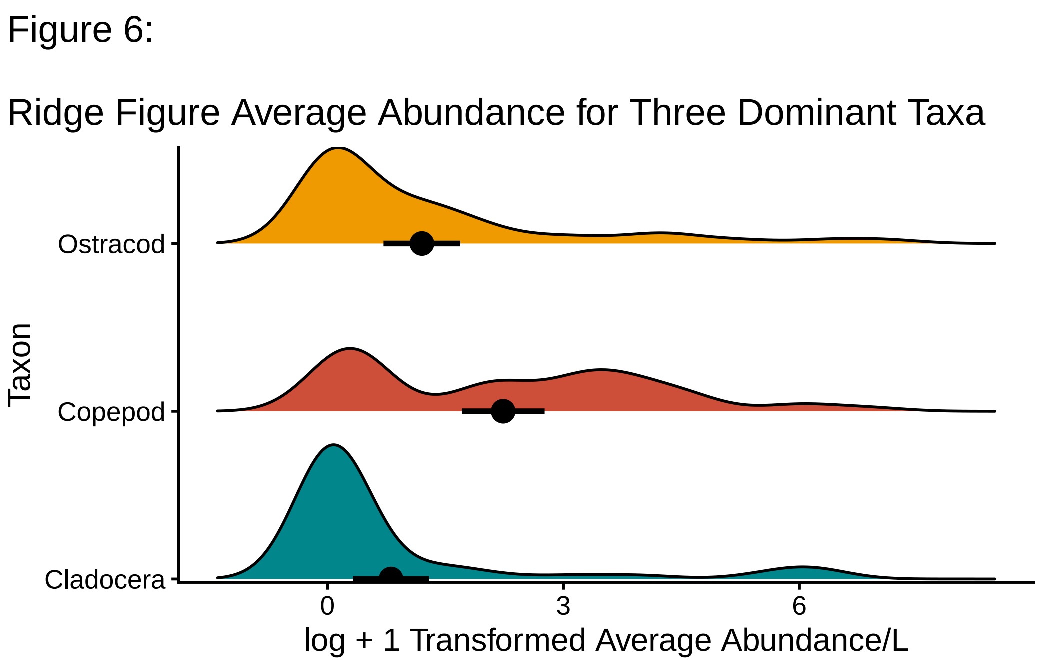


Figure 6 - How is abundance distributed across the three dominant taxa?

```
# creating new object for ridgeline
figure6 <- ggplot(data = aq_ins_clean,
                  # x-axis: avg temp
                  # y-axis: avg abund
                  # fill geometries by site
                  mapping = aes(x = log_ave_lit,
                                y = taxon,
                                fill = taxon)) +
  # ridges (scale makes ridges shorter so that they don't overlap)
  geom_density_ridges(scale = 0.8) +

  # adding means and 95% confidence intervals
  stat_summary(geom = "pointrange",
               fun.data = mean_cl_normal,
               size = 0.75,
               linewidth = 1) +
  # filling geometries with color
  scale_fill_manual(values = taxon_cols) +
```

```
  # setting y-axis expansions
  scale_y_discrete(expand = expansion(mult = c(0.01, 0.15))) +
  # relabelling x- and y-axis
labs(title = "Figure 6:

Ridge Figure Average Abundance for Three Dominant Taxa",
       y = "Taxon",
       x = "log + 1 Transformed Average Abundance/L")

# displaying object
figure6
```



Ideas of Analysis for Background

Background about NCOS and project history:

From 1966 to 2013, the space where NCOS sits was used as a 9-hole golf course. After a couple years of extensive planning, the NCOS restoration project began in February 2017, managed by the UCSB Cheadle Center. This space has created more than 40 acres of estuarine and palustrine wetlands and restored more than 60 acres of upland habitats. In comparison with COPR, the study has found that NCOS appears to have equivalent, if not slightly greater species richness and evenness.

Specifics about aquatic invertebrate:

Samples are collected by student interns and volunteers using a filtered bucket and dip net method, with the 3 most common invertebrates found with the filtered beaker 250 micron method being copepods, cladocera, ostracods and the 3 most common invertebrates found with the dipnet 500 micron method being copepods, ostracods, hemipteras. There is a strong negative linear relationship between aquatic invertebrate and salinity, as shown by a 2022 monitoring report of Devereux slough. Additionally found in this report, "from the data so far, the Slough and its associated ponds and vernal pools contain a fairly small set of invertebrate inhabitants. While we will need more time to determine what NCOS's steady-state environment will be like, COPR's portion of the Slough has some relatively extreme conditions with regard to Salinity, Temperature, and Dissolved Oxygen". Only four planktonic taxa appear in any great abundance: Copepod, Corixidae, Ostracod, and Cladocera. The benthic subsrate has a much higher concentration of these taxa, which is where the dip net testing is now most focusing on. In the 2024 Aquatic Invertebrate Assessment at the North Campus Open Space, it was noted that the highest species diversity was found at the Phelp's creek site compared to a couple other NCOS sites. There are other reports worth exploring that compare the linkage between aquatic invertebrate and other animals, such as waterbirds.

Important Questions and relationships that can be explored:

- What is the relationship between aquatic invertebrates and temperature?
- Does abundance vary across site? How does abundance vary across temperature ranges for each taxon? (exploratory)
- How does abundance vary across sites for each taxon? (exploratory)
- What is the strength of the relationship between water temperature and invertebrate abundance, and does this differ by season?
- How is abundance distributed across the three dominant taxa? (exploratory)

Links to reports:

- https://escholarship.org/uc/item/16p046mb#main
- https://escholarship.org/uc/item/59c872mm
- https://escholarship.org/uc/item/64f0w6hx
- https://escholarship.org/uc/item/6n78q34g
- https://escholarship.org/uc/item/1bc3t7fb

Plan for elective

Timeline

Week 7:

- Research all three organisms (copepod, ostracod, cladoceran)
- Gather reference images

- Review your figures
- Take notes on what each figure shows about the species
- Brainstorm collage/drawing style and decide on materials (digital vs. physical).

Week 8 (timeline check-in):

- Have a rough sketch/draft of all three 2D collages– one per organism
- Bring your notes on what the figures show about each species. This is our plan and draft milestone.

Week 9:

- Refine and finalize the artwork for 1–2 of the organisms
- Fill in your species notes with complete observations from your figures.

Week 10:

- Complete the third organism's artwork and finish all species notes
- Do a full review, make sure each collage clearly reflects what your figures show about that species.

Finals week:

- Final polish, compile everything together
- Present your completed 2D collage triptych with notes.

Visualization Approach and Temperature Color Scale

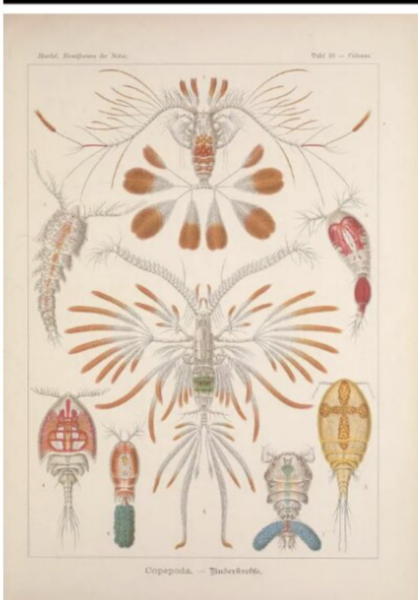
For our final visualization, we decided on a physical drawing. Each organism will be represented by an outline, filled with paint and pastels to illustrate the relationship between abundance and temperature. The colors inside each outline correspond to a temperature-based heat range, divided into five equal intervals spanning the full range of recorded temperatures (10.20°C – 37.30°C):

- 10.0°C – 15.6°C
- 15.6°C – 21.2°C
- 21.2°C – 26.8°C
- 26.8°C – 32.4°C
- 32.4°C – 38.0°C

Gathering Images and Research on Organisms

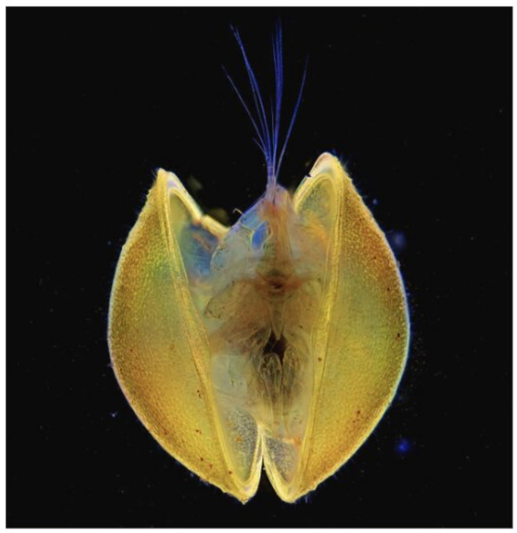
Copepods

- Group of small crustaceans
- Found in nearly every salt and freshwater habitat
- Some are planktonic, some benthic (live on sediments)
- Teardrop shaped body with large antennae
- Thin exoskeleton is transparent; they eat phytoplankton, detritus, and small zooplankton
- Several species are bioluminescent
- They jump out of the water to evade predators



Ostracods

- tiny, ubiquitous crustaceans ranging from 0.1 to 30 mm in size
- Commonly known as “seed shrimp”
- Bivalve shells that encase their bodies
- About 33,000 species, some being bioluminescent
- Most common arthropods in the fossil record



Cladocera

- Commonly known as the water flea
- Freshwater crustaceans, feeding on microscopic organics, though some forms are predatory
- Over 1000 species
- Down-turned head with a single median compound eye, and a carapace covering the apparently unsegmented thorax and abdomen
- Most cladoceran species live in fresh water and other inland water bodies